

# Positive Literature Answer to the Canonical Orbit-Pseudometric Question in arXiv:1804.11164

FA/Banach run packet

2026-07-18

## Status

This is a literature-already-answered packet. It records that a question from the concluding problems of Cuth–Doucha–Kurka [1] was answered positively in the later paper of Kurka [2]. It is not an original proof from this run.

## Original Problem

In the concluding remarks of [1], the authors ask for a continuous version of the Clemens–Gao–Kechris theorem. Let  $d$  be the Hausdorff distance on  $F(\mathbb{U})$ , let  $\text{Iso}(\mathbb{U})$  act canonically on  $F(\mathbb{U})$ , and let  $\rho$  be the corresponding orbit pseudometric. The question is whether

$$\rho_{GH} \quad \text{and} \quad \rho$$

are Borel-uniformly continuously bi-reducible, or whether at least one of the two reductions holds.

## Supporting Answer

Kurka [2, Theorem 1.2] proves precisely that

$$\rho_{GH} \equiv_{B,u} \rho_{\text{Iso}(\mathbb{U}), \rho_H}$$

on  $F(\mathbb{U}) \setminus \{\emptyset\}$ , where  $\rho_H$  is the Hausdorff distance. This is the same canonical Urysohn-space orbit pseudometric asked about in [1], restricted to the usual nonempty closed-set coding of Polish metric spaces.

The same paper proves a stronger comparison theorem. In Theorem 1.3 it constructs a Borel map  $p \mapsto Y_p$  such that

$$\rho_{GH}(Y_p, Y_q) \leq \rho_{\text{Iso}(\mathbb{U}), \rho_H}(Y_p, Y_q) \leq \rho_{G,d}(p, q) \leq 2\rho_{GH}(Y_p, Y_q).$$

Consequently the reductions in Theorem 1.2 can be chosen Borel-Lipschitz on small distances.

**Literature Answer.** *The canonical orbit-pseudometric question in [1] has a positive literature answer: the Gromov–Hausdorff pseudometric is Borel-uniformly continuously bi-reducible with the orbit pseudometric induced by the canonical action of  $\text{Iso}(\mathbb{U})$  on  $F(\mathbb{U})$  equipped with Hausdorff distance.*

*Literature identification.* The question in [1] uses exactly the data  $(F(\mathbb{U}), \text{Iso}(\mathbb{U}), \rho_H)$ . Kurka’s Theorem 1.2 states the Borel-u.c. bireducibility of  $\rho_{GH}$  and  $\rho_{\text{Iso}(\mathbb{U}), \rho_H}$  for this data. Therefore both directions requested by the question are supplied by the later paper.  $\square$

## Scope Limitations

This packet answers only the specific canonical Urysohn/Hausdorff orbit-pseudometric question. It does not solve the other concluding problems from [1]; in particular, Kurka leaves open a broader question about removing the auxiliary system assumption in the general orbit-pseudometric reduction theorem.

## Verification Notes

The source and supporting papers are distinct. The supporting paper cites [1] and explicitly presents Theorem 1.2 as a positive answer to Question 41 from that paper. The local duplicate search covered arXiv ids 1804.11164 and 2204.08375, plus phrases `Gromov-Hausdorff`, `orbit pseudometric`, `Iso(U)`, `Hausdorff distance`, and `bireducible`.

## References

- [1] M. Cuth, M. Doucha, and O. Kurka, *Complexity of distances: Theory of generalized analytic equivalence relations*, arXiv:1804.11164; J. Math. Log. 22 (2022), no. 1, 2250005.
- [2] O. Kurka, *Orbit pseudometrics and a universality property of the Gromov-Hausdorff distance*, arXiv:2204.08375; Topology Appl. 364 (2025), 109095.